Properties of R_j

$$R_j = \frac{1}{2}(P_j - iQ_j) \qquad j \neq 0.$$

$$\begin{aligned}
\mathbb{E}(R_j) &= \mathbb{E}\left[\frac{1}{2}(P_j - iQ_j)\right] \\
&= \frac{1}{2}\mathbb{E}(P_j) - \frac{i}{2}\mathbb{E}(Q_j) \\
&= 0 \qquad \because \mathbb{E}(P_j) = 0 \text{ and } \mathbb{E}(Q_j) = 0.
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}\left[|R_j|^2\right] &= \mathbb{E}\left[R_j \overline{R_j}\right] \\
&= \mathbb{E}\left[\frac{1}{2}(P_j - iQ_j) \times \frac{1}{2}(P_j + iQ_j)\right] \\
&= \frac{1}{4}\mathbb{E}\left[P_j^2 + Q_j^2\right] \\
&= \frac{1}{4}\left[\mathbb{E}(P_j^2) + \mathbb{E}(Q_j^2)\right] \\
&= \frac{1}{4}(2g_j + 2g_j) \\
&= g_j.
\end{aligned}$$